C Kaustubh

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Education

B.Tech in Computer Science & Engineering at PES University 2023 – 2027
CGPA: 8.56 **Dr. CNR Rao Scholarship** (Sem 1) **DAC Distinction Scholarship** (Sem 2 – Sem 5)

Experience

Research Intern @ **Center for Information Security, Forensics and Cyber Resilience (C-ISFCR)** Jun '25 – Jul '25
Under Prof. Preet Kanwal

- Designed and evaluated four multi-modal architectures (YOLOv11 + Optical Flow + LLaVA-Next + LLaMA-3) for unsupervised video anomaly detection, achieving **75.48% classification accuracy** on the NWPU Campus Dataset across 16.5 hours of footage.
- Developed a novel keyword-free composite anomaly scoring framework combining semantic (SBERT), visual (ResNet-50), object distribution (Jensen-Shannon divergence), and temporal dynamics signals; F1 scores of **0.845 – 0.895** across all test categories.
- Built a three-tier adaptive summarization engine supporting arbitrary-length video via hierarchical divide-and-conquer LLM summarization; GPU-accelerated inference on RTX 4090 with CUDA 12.1, PyTorch 2.1, and FFmpeg achieving real-time frame processing.
- Benchmarked architecture against Gemini-2.5-Flash on five NWPU context anomaly videos; comparable anomaly scores with significantly lower serving cost.

Leadership & Activities

Tech Lead @ Shunya PESU Jan '26 – Present
Head of Sponsorship @ Nexus PESU : Organised NexGen 1.0 & 2.0; raised **INR 4L+** in sponsorships. Jan '24 – Jul '25
Board of Directors @ ECell PESU : Conducted E-Summit 2024. Feb '24 – Jul '25

Skills & Technologies

Languages & Frameworks: Python, Rust, C/C++, Go, TypeScript, Next.js, PyTorch, Bash
Infrastructure & DevOps: Docker, Kubernetes, GitHub Actions (CI/CD), Linux, Git, Redis, Kafka, Spark
ML & Data: PyTorch, CUDA, OpenCV, Sentence-BERT, YOLOv11, LLaVA, LLaMA, Flower (Federated Learning)

Projects

- **Rakshak-AI** — An agentic compliance readiness copilot that helps Indian startups and regulated companies turn scattered security evidence into audit-ready answers for DPDP, CERT-In, RBI, and ISO 27001. Fully deployed on GCP. 1st Runner-up at **GDG Cloud Bengaluru Deployfest 2026**
- **Qwen_Cuda** — Raw CUDA inference for the Qwen Model family. Full hardware accelaration with custom kernels. Beats transformers on Qwen3:8B Q4_K by +3 tok/s.
- **Docksmith** — A simplified Docker-like build and runtime system built from scratch in Go.
- **Orderflow** — A real-time market data pipeline that simulates high-frequency trading events and turns PostgreSQL CDC streams into Kafka-powered Spark analytics (VWAP, imbalance, volatility, and whale alerts) at production-style scale. CI via GitHub Actions.
- **UncertaRAG** — Uncertainty-aware RAG evaluation framework benchmarking Mistral-7B across 12 quantization conditions (FP16 vs INT4). Key result: a no-retrieval SemanticVar baseline outperforms all retrieval-augmented variants on AUROC (0.815 under INT4).
- **redis-from-scratch** — Redis-compatible in-memory key-value store in Go with full RESP3 protocol support and Pub-Sub. Deployed as backing store for a REST API-based Kafka clone. CI via GitHub Actions.
- **sql-engine** — An toy SQL database engine implementation from scratch in C++ to understand database internals. CI via GitHub Actions.
- **CodeRonin** — AI-powered Python debugging game where an agent injects realistic bugs to build library and debugging mastery. **Agentathon '26 Winner**.